

Numerical Analysis PhD Exam, May 2010

Answer any 8 questions.

1. If P is a projector that is neither 0 nor the identity, prove that $\|P\| = \|I - P\|$ in any norm induced by a vector norm generated by an inner product.
2. Let $M = \begin{bmatrix} A & B^t \\ B & C \end{bmatrix}$ be a real symmetric and positive definite matrix with an invertible A . Prove that the Schur complement $S = C - BA^{-1}B^t$ is also symmetric and positive definite.
3. Suppose A is a Hermitian positive definite matrix split into $A = C + C^* + D$ where D is also Hermitian positive definite. Prove that $B = C + \omega^{-1}D$ is invertible whenever $0 < \omega < 2$. Consider the iteration $x_{n+1} = x_n + B^{-1}(b - Ax_n)$, with any initial iterate x_0 . Prove that x_n converges to $x = A^{-1}b$ whenever $0 < \omega < 2$.
4. If a square matrix A can be block partitioned as $A = \begin{bmatrix} 0 & M \\ N & 0 \end{bmatrix}$, prove that $\sigma(A) = \{z \in \mathbb{C} : z^2 \in \sigma(MN) \cup \sigma(NM)\}$.
5. Suppose A is a Hermitian positive definite matrix with eigenvalues

$$0 < \lambda_0 \ll \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N.$$

Let $\kappa = \lambda_N/\lambda_0$ and $\kappa' = \lambda_N/\lambda_1$. Let e_n denote the error in the n th step of the conjugate gradient algorithm applied to $Ax = b$, and $\|y\|_A = \langle Ay, y \rangle^{1/2}$. Prove that

$$\|e_n\|_A \leq 2\kappa \left(\frac{\sqrt{\kappa'} - 1}{\sqrt{\kappa'} + 1} \right)^{n-1} \|e_0\|_A.$$

6. Let n be a positive integer, $h = 1/n$, and consider the grid of points (ih, jh) for $i, j = 0, 1, \dots, n$. Let A be the finite difference operator with the “5-point stencil” discretizing the Laplace operator $-\Delta$, with zero Dirichlet boundary conditions on this grid. Describe it. What are the eigenvalues of A ? Prove that the spectral condition number of A is $\kappa(A) = \cot^2(\pi h/2)$.
7. Let $x_0, x_1, \dots, x_n$ be distinct points in a finite interval $[a, b]$ and $f \in C^1[a, b]$. Show that for any given $\varepsilon > 0$ there exists a polynomial p such that $\|f - p\|_\infty < \varepsilon$ and $p(x_i) = f(x_i)$ for all $i = 0, 1, \dots, n$ (where $\|\cdot\|_\infty$ denotes the $L^\infty(a, b)$ -norm).
8. Let $w(x)$ be a positive integrable function on $[a, b]$. Consider the quadrature

$$\int_a^b f(x)w(x) dx \approx \sum_{k=0}^n (A_k f(x_k) + B_k f'(x_k) + C_k f''(x_k))$$

for any f with continuous first and second derivatives (f' and f'' , respectively). Give conditions on A_k, B_k, C_k and x_k so that the quadrature has precision $4n + 3$.

9. Given that the Newton iteration for finding a root r of f converges, $f \in C^2(\mathbb{R})$, and $f(r) = f'(r) = 0 \neq f''(r)$, prove that the convergence cannot be quadratic. Suggest a modification that restores quadratic convergence for smooth f .

10. Let $x_0, x_1, \dots, x_n$ be distinct numbers and for each i , let ℓ_i denote the product of all $(x_i - x_j)^{-1}$ for all $j \neq i, j = 0, \dots, n$. Express the number $q_f = f(x_0)\ell_0 + f(x_1)\ell_1 + \dots + f(x_n)\ell_n$ as a divided difference of f . Then show that $q_f = 0$ whenever f is a polynomial of degree $n - 1$.